

BIT (Math 2041 Worksheet III)

1. For each of the following curves; find the arc length, unit tangent vector, unit normal vector and curvature:
 - a) $\mathbf{C}: \mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 4t \rangle; 0 \leq t \leq \pi$
 - b) $\mathbf{C}: \mathbf{r}(t) = (1 - 5 \cos t)\mathbf{i} + (5 \sin t)\mathbf{j} - 12t\mathbf{k}; 0 \leq t \leq \pi$.
2. Evaluate the following line integrals:
 - a) $\int_C (x^3 + y^3) ds; \mathbf{C}: \mathbf{r}(t) = \langle 3e^{2t} + 1, 4e^{2t} + 2 \rangle, 0 \leq t \leq \ln 2$
 - b) $\int_C (2x - y) ds; \mathbf{C}: \mathbf{r}(t) = \cos^3 t \mathbf{i} + \sin^3 t \mathbf{j}; 0 \leq t \leq \pi$
 - c) $\int_C x^3 y^4 dx + x^4 y^3 dy; \mathbf{C}: \mathbf{r}(t) = \langle \sqrt{t}, 1 + t^3 \rangle, 0 \leq t \leq 1$
 - d) $\int_C (6y - \sec x) dx + (2x - e^{\cot y}) dy; \mathbf{C}$ is the boundary of the region between the graphs of $y = x$ and $y = x^2 - 2x$.
 - e) $\int_C 15xy dx + (4x^3 - 2y) dy; \mathbf{C} = \mathbf{C}_1 \cup \mathbf{C}_2$ where $\mathbf{C}_1$ is the graph of $y = x^3$ from $(0,0)$ to $(1,1)$ and $\mathbf{C}_2$ is the line segment from $(1,1)$ to $(2,4)$.
3. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where
 - a) $\mathbf{F} = \langle y, -x, xyz^2 \rangle$ and $\mathbf{C}: \mathbf{r}(t) = \langle e^{-t}, e^t, t \rangle, 0 \leq t \leq 1$
 - b) $\mathbf{F} = (e^y + ye^x)\mathbf{i} + (xe^y + e^x)\mathbf{j}$ and $\mathbf{C}: \mathbf{r}(t) = \sin(\pi t)\mathbf{i} + (\ln t)\mathbf{j}, 1 \leq t \leq 2$
 - c) $\mathbf{F} = \frac{8x^3}{1+x^4+y^4+z^4}\mathbf{i} + \frac{8y^3}{1+x^4+y^4+z^4}\mathbf{j} + \frac{8z^3}{1+x^4+y^4+z^4}\mathbf{k}$ and $\mathbf{C}: \mathbf{r}(t) = t^2\mathbf{i} + t^4\mathbf{j} + t^6\mathbf{k}, 0 \leq t \leq 1$
 - d) $\mathbf{F} = 5 \ln(6 + y)\mathbf{i} - \frac{xy+x}{6+y}\mathbf{j}$ and $\mathbf{C}$ is the triangle with vertices $(0,0)$, $(2,0)$, and $(0,4)$.
 - e) $\mathbf{F} = 2z^2\mathbf{i} + e^y \ln y \mathbf{j} + 3y^2\mathbf{k}$ and $\mathbf{C}$ is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $2x + y + z = 4$ oriented positively.
 - f) $\mathbf{F} = (x^2 + z)\mathbf{i} + (x + y^2)\mathbf{j} + (y + z^2)\mathbf{k}$ and $\mathbf{C}$ is the intersection of the sphere $x^2 + y^2 + z^2 = 8$ and the cone $z = \sqrt{x^2 + y^2}$ oriented counterclockwise.
4. Compute the surface integrals $\iint_S \mathbf{g}(x, y, z) dS$.
 - a) $\mathbf{g}(x, y, z) = x^2 + y^2$ and S is the upper half of the sphere $x^2 + y^2 + z^2 = 1$.
 - b) $\mathbf{g}(x, y, z) = \sqrt{1 + 4z}$ and S is the part of the paraboloid $z = x^2 + y^2$ inside the cylinder $x^2 + y^2 = 2$
5. Evaluate the following surface integrals.
 - a) $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS; \mathbf{F} = z\mathbf{i} + x\mathbf{j} + xyz\mathbf{k}$ and S is the top half of the unit sphere $x^2 + y^2 + z^2 = 1$.
 - b) $\iint_S \nabla \mathbf{g} \cdot \mathbf{n} dS; g = xe^y \sin z$ and S is the sphere $x^2 + y^2 + z^2 = 4$.
 - c) $\iint_S \mathbf{F} \cdot \mathbf{n} dS; \mathbf{F} = 2x^4\mathbf{i} + 3xy^3\mathbf{j} - xy^2z\mathbf{k}$ and S is the surface of the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0, z = x$.
 - d) $\iint_S \mathbf{F} \cdot \mathbf{n} dS; \mathbf{F} = \tan(y^2)\mathbf{i} + (x^4 + 8yz)\mathbf{j} - 3z^2\mathbf{k}$ and S is the surface of the solid inside the sphere $x^2 + y^2 + z^2 = 2z$ and above the cone $z = \sqrt{x^2 + y^2}$.

BIT (Math 2041 Worksheet IV)

1. Show that the following functions are differentiable and find $f'(z)$.
a) $f(z) = e^{2x} \cos 2y + (e^{2x} \sin 2y)i$ b) $f(z) = x^3 - 3xy^2 + (3x^2y - y^3)i$
2. Check whether the following functions are harmonic or not. For these which are harmonic, find the corresponding harmonic conjugate and analytic function $f(z)$.
a) $u(x, y) = x^2 - y^2 + 7x + 3y$ c) $u(x, y) = e^x \cos y - e^y \cos x + xy$
b) $u(x, y) = x^4 - 6x^2y^2 + y^4 + x$ d) $u(x, y) = \ln(x^2 + y^2)$
3. Evaluate the following counter integrals.
a) $\int_C (x^3 - y^3)dz$ where C is the lower half of the circle $|z| = 1$ from $z = -1$ to $z = 1$.
b) $\int_C \left(\frac{1}{(z+i)^3} - \frac{5}{z} \right) dz$ where C is the circle $|z + i| = 1$.
c) $\int_C f(z)dz$ where C is the arc from $z = -1 - i$ to $z = 1 + i$ along $y = x^3$ and
$$f(z) = \begin{cases} 1, & y < 0 \\ 4y, & y > 0 \end{cases}$$
4. Evaluate the following integrals using Residue Theorem.
a) $\int_C \frac{e^z}{z(z-1)(z^4-16)} dz; C: |z - i| = 2$ d) $\int_C \frac{z^2-2z}{(z-1)^2(z^2+1)} dz; C: |z + i| = 3$
b) $\int_C \frac{\cos(\pi z)}{(z^2-z)(z+i)^2} dz; C: |z + i| = \frac{3}{2}$ e) $\int_C \frac{\cos 3z}{z^5} dz; C: |z| = 1$
c) $\int_C (z+2)^2 \sin\left(\frac{1}{z+1}\right) dz; C: |z| = 3$
5. Evaluate the following integrals using Residue theorem.
a) $\int_0^{2\pi} \frac{1}{3+\cos^2\theta} d\theta$ b) $\int_0^{2\pi} \frac{\cos\theta}{3+\sin\theta} d\theta$ c) $\int_{-\infty}^{\infty} \frac{dx}{(x+1)(x^2+2)}$ d) $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+9)^2}$